

Artifact Suppression for Passive Cavitation Imaging Using U-Net CNNs with Uncertainty Quantification

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Abstract—The clinical applications of chemotherapy treatments for brain tumors are limited by the blood-brain barrier (BBB) which blocks drugs from reaching the tumors. Magnetic resonance imaging (MRI)-guided focused ultrasound (FUS) is applied to the brain to create micro-bubbles in the BBB for opening the BBB temporarily for targeted drug delivery. Creation of these microbubbles is monitored using passive cavitation imaging (PCI). Generally, PCI utilizes delay-and-sum (DAS) beam-forming, which suffers from a space-varying point spread function and high sidelobes both of which introduce image artifacts. To address this issue and to better monitor the BBB opening, a deep learning denoising approach is proposed based on a U-Net convolutional neural networks (CNN) architecture. The U-Net is trained with pairs of images, with DAS passive cavitation images at the input layer, and ground true noise-free images at the output layer. In the training phase, the network fits the mapping from the DAS images to the noise-free images. A Monte Carlo (MC) dropout approach is used to quantify uncertainty in de-noised images, giving information that can be further used to suppress artifacts. It is shown in this work that the resulting model can generate de-noised DAS images, and that uncertainty quantification in U-net output can be used to further improve artifact suppression.

Keywords—neural networks; source localization; image denoising; focused ultrasound; trans-cranial therapy; uncertainty quantification

I. INTRODUCTION

Clinical chemotherapy for brain disease is severely limited by the blood-brain barrier (BBB) which prevents cytotoxic levels of compounds from entering the brain. While protective in general, the BBB complicates therapy by blocking drugs from reaching malignant tumor cells in the brain. Current research indicates that temporarily disrupting the BBB at the tumor location by focused ultrasound (FUS) is a potential non-invasive solution to this problem [1]. Its

uses in the brain, however, are limited by the skull as well as by the difficulty of targeting and monitoring treatment [2]. While the attenuation and distortion of the ultrasound field introduced by skull bone pose a serious challenge [3], recent work has shown that trans-cranial therapy is possible and that high intensity focused ultrasound (HIFU) can be developed for brain therapy. Specifically, MRI and Computed Tomography (CT) can be used as the basis for computing the corrections needed to adjust the delivered focused ultrasound (US) for the effects of the skull [3, 4].

While MRI and CT can be used to help deliver the US to the brain, these modalities are not as useful for quantifying the bio-effects. Indeed, the blood-brain barrier opening is caused by cavitation, which is the oscillation of bubbles in response to external stimuli such as focused ultrasound [5]. Cavitation can be monitored noninvasively by passively receiving the acoustic field introduced by the bubble activities using a matrix of transducers. Passive cavitation imaging (PCI, also known as passive acoustic mapping) uses this data to estimate positions of cavitation [6, 7], usually based on the delay-and-sum algorithm. Since properly delayed signals from sources add up by constructively, image pixel amplitudes are related to the intensities of the acoustic sources.

Recent years have seen application of adaptive beamforming to PCI [8, 9], as well as increasing interest in applying deep learning (DL) methods to image formation [10, 11]. In the context of X-ray computed tomography, DL-based reconstruction can lead to false image features which can be corrected using Bayesian methods [12].

In this paper, we first briefly review delay-and-sum (DAS)-based cavitation imaging, then introduce the U-Net based Deep Learning approach to artifact suppression. We then show the estimated uncertainty in U-Net outputs and propose methods for using uncertainty quantification to suppress spurious sources. This work is similar in flavor to [12] but utilizes easily-implemented Monte Carlo dropout

methods [13] as well as PCI-specific statistical tests. The main contributions of the work are that it represents, to our knowledge, the first application of DL-based methods to PCI imaging, and that we propose novel methods for using uncertainty quantification information to suppress spurious sources.

II. BACKGROUND

Usually, the passive cavitation imaging model assumes free-space propagation without boundaries (no reflection or refraction), and constant medium density ρ and wave speed c (so travel times are dependent only on distances). Assuming the point source at $\vec{y}$ radiates acoustic waves with pressure $p(t)$, the spherical wave propagating outward from the source can be represented as the convolution of $p(t)$ with the free-space time-domain Green's function [14]. Therefore, the signal s at an observation point $\vec{x}$ is given by,

$$s(t, \vec{x}) = p(t) * G(\vec{x}, \vec{y}, t) = \frac{p(t - \frac{|\vec{x} - \vec{y}|}{c})}{4\pi|\vec{x} - \vec{y}|}. \quad (1)$$

Beside additive noise, images generated by DAS-based PCI typically have extended axial “tail” artifacts due to the point spread function (PSF) [15, 16]. Mathematically, the object plane field, can be represented as

$$O(x_o, y_o) = \iint O(u, v) \delta(x_o - u, y_o - v) du dv, \quad (2)$$

where $\delta(x_o - u, y_o - v)$ denotes the impulse at the point source location, and $O(u, v)$ denotes the source intensity at (u, v) . The resulting DAS image $I(x_o, y_o)$ is expressed as

$$I(x_o, y_o) = \iint O(u, v) \text{PSF}(x_o, y_o, u, v) du dv, \quad (3)$$

where $\text{PSF}(x_o, y_o, u, v)$ denotes the point spread function, or image of the impulse function $\delta(x_o - u, y_o - v)$. As seen in Eq. (3), the PSF is in general a function of the point source location.

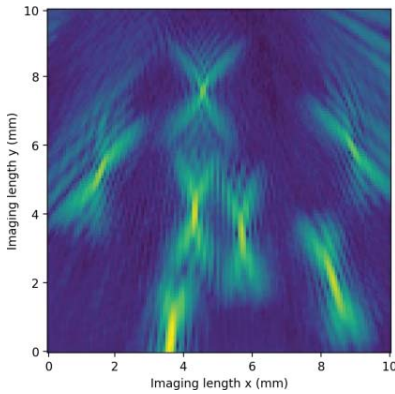


Figure 1. Simulated passive cavitation imagery. The PCI generation parameters are listed in Table 2.

Superposition applies to PSF-induced artifacts [8]. When bubbles are intensively distributed in a region, the image of a bubble may overlap with that of the others, making point source locations in the image difficult to distinguish. Fig. 1 shows a simulated DAS imagery of 7 micro-bubbles on log-scale with 30dB of dynamic range.

DAS beamforming is used for directional transmission and reception. Fig. 2 is a schematic illustration of a point source acoustic emission and its delay-and-sum beamforming. It is achieved by arranging the data from transducer elements in a receiver matrix in such a way that acoustic signals from specific directions have constructive interference whereas others have destructive interferences [17]. The received signals are separately delayed according to the computed travel time from the detection point to the transducers. Constructive interference would be found only if the delays are corresponding to the traveling time from the actual acoustic source to the transducers. Fig. 2 illustrates the constructive and destructive interferences of recorded pulses from a point source emission, under the delay-and-sum mechanism.

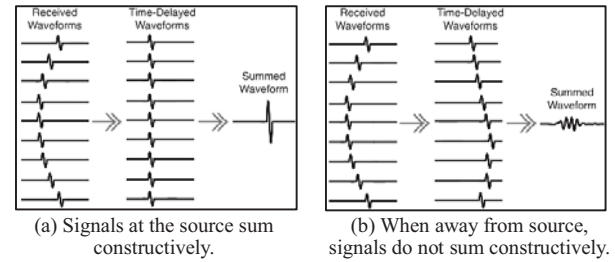


Figure 2. Schematic illustration of the delay-and-sum mechanism. (a) The waveforms sum constructively at the point source location. (b) Away from the point source location, the waveforms do not sum constructively and results in forming lower magnitude and less energy. These plots are modified from Haworth et al. IEEE Transactions on Ultrasonics, Ferroelectrics and Frequency Control, 2016, doi: 10.1109/TUFFC.2016.2620492; Copyright © 2016 IEEE.

III. U-NET BASED LOCALIZATION

The U-Net architecture was first proposed for medical image segmentation and cell tracking [18]. The U-Net utilizes skip connections in its architecture to facilitate the learning [19]. Long and short skip connections have been shown to be helpful in recovering spatial resolution [20] and speeding up convergence [10], respectively. Subsequently the U-Net was applied to medical imaging inverse problems [10, 11].

In this work, several changes are made to the original U-Net [18] to adapt it for use in DAS artifact suppression. Since the U-Net in this work is used for de-noising while the original U-Net is used for segmentation tasks, the loss function is changed from binary cross-entropy to mean-squared error, and activation function at the output layer are changed from rectified linear unit to linear. Nonlinearity is introduced by convolutions in the hidden layers. Importantly, Monte Carlo dropout is added to the U-Net for producing variational output [13]. In this approach, the network is trained using dropout, and dropout is also used during prediction to give a statistical sampling of U-net outputs.

Fig. 3 shows the modified U-Net architecture used in this work. Table 1 shows more details about the configuration. Parameters are based on the original U-Net [18] with the changes discussed above, and additional dropout applied for later variational inferences.

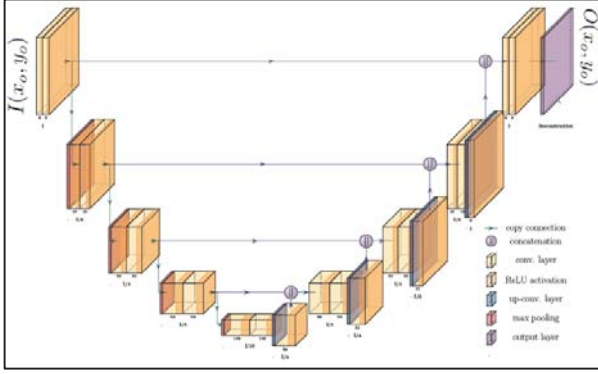


Figure 3. U-Net architecture. DAS images (noted as $I(x_o, y_o)$, same as in (3)) in the training set are input to the U-Net (upper left corner), the corresponding multiple single-pixel dots (noted as $O(x_o, y_o)$, same as in (2)) indicating ground true point source locations are set at output layer (upper right corner) while training the U-Net.

TABLE I. U-NET CONFIGURATIONS

	Configuration name	values
Convolutional layers	Convolution size	3×3
	Convolution stride	2×2
	Padding	Same
	Deconvolution size	2×2
	Deconvolution stride	2×2
	Max pooling size	2×2
	Dropout rate	10%

IV. EXPERIMENTS

A. Data Simulation

In the simulation, a set of randomly located acoustic sources are modeled, and signals from each source are time-delayed based on the distances from sources to receivers on the receiver matrix. DAS imaging is then applied to the simulated data. Source intensities and corresponding ground-truth image intensities vary between 0.5 to 1 (linear range). Table 2 shows details of the PCI simulation setup. Here, the imaging area is 10mm×10mm for artifact suppression demonstration purposes. However, future works could consider a larger imaging area.

TABLE II. PCI SIMULATION SETUP

Simulation setup	values
Frequency	40MHz
Time of capturing	10ms
DAS averaging window length	2ms
Noise type	Gaussian
Signal-to-noise ratio (linear)	10
Source acoustic intensity	0.5-1
Number of micro-bubbles (random)	1-10
Image size (pixels)	128×128
Imaging area (mm ²)	10×10

B. Results from Averaged U-Net Output

Fig. 4 demonstrates trained U-Net model's performance on six randomly drawn DAS images. The images shown are averages of 100 U-Net outputs with Monte Carlo (MC) dropout. The DAS images (left) are input into the U-Net and used to generate denoised images (right). The model works well as localizing almost all point sources in the noisy DAS image, and the U-Net reduces background levels very noticeably. However, artifacts in the DAS images lead to false positive or "false" sources in the U-Net outputs.

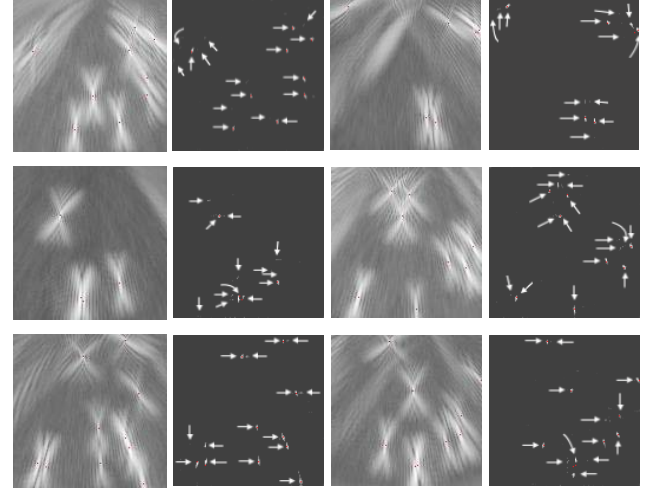
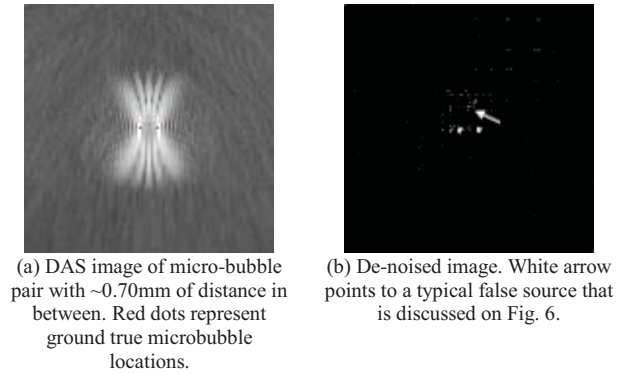
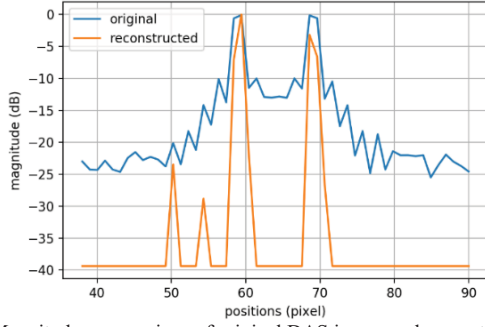


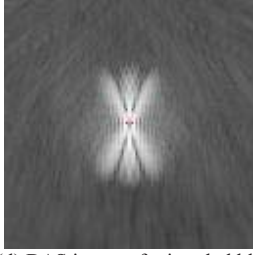
Figure 4. Testing results. DAS images of multiple micro-bubbles (column 1 and 3), vs. the corresponding U-Net outputs (column 2 and 4), on log-scale. All reconstructed sources are pointed by white arrows. Red dots denote ground true point source locations, thus, sources with red dot on them are real sources, otherwise false sources.

In simulation, we can control the distances between two point-sources, letting us study how performance depends on source geometry. Results on log-scale of a lateral test on the distinguishability of micro-bubbles and the performance of denoising under the trained U-Net are presented in Fig. 5. It shows that false sources may be generated if the true acoustic sources are close. The magnitude of false sources is seen to increase when the true acoustic sources move closer to each other (Fig 5d-e).

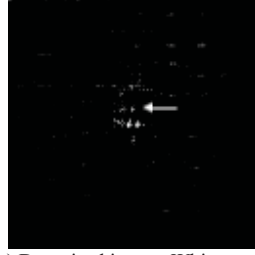




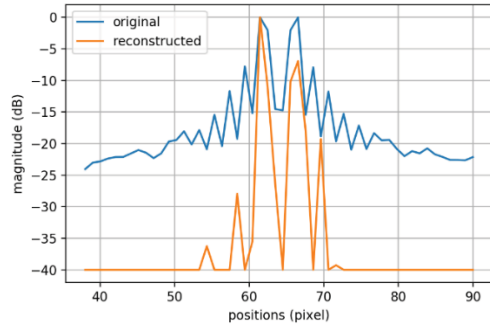
(c) Magnitudes comparison of original DAS image and reconstructed image at line cut through the microbubble pair shown in Fig. 5(a) and (b).



(d) DAS image of micro-bubble pair with ~0.23mm of distance in between. Red dots represent ground true microbubble locations.



(e) De-noised image. White arrow points to a typical false source that is discussed on Fig. 6.



(f) Magnitudes comparison of original DAS image and reconstructed image at line cut through the microbubble pair shown in Fig. 5(d) and (e).

Figure 5. Lateral resolution test on two closely distributed microbubble pairs.

C. Uncertainty Quantization

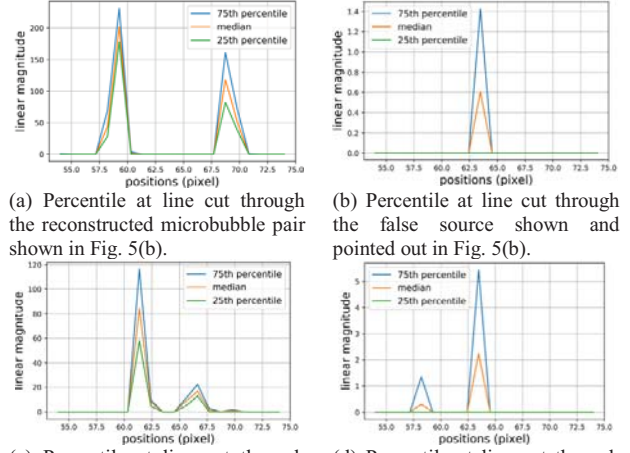
In this work, 100 reconstructions were generated for every DAS input using MC dropout. Fig. 6 explores the variability of the pixel-wise magnitudes for both real and false sources. It is observed that reconstructed real source values are stable and are higher than those of false sources.

A first, simple percentile filtering model $\mathcal{F}$ was proposed here for distinguishing real and false sources. Mathematically, $\mathcal{F}$ can be expressed as

$$\mathcal{F}(I) = \begin{cases} I, & P_{15} > 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where I denotes the pixel-wise image data and P_{15} denotes the 15th percentile. Fig. 7 shows the results of applying this

filter to the examples shown in Fig. 5(b) and 5(e). In Fig. 7, almost all false sources are filtered out by the filter $\mathcal{F}$, which means that false sources' distributions are densely distributed near 0 and rarely have large values. In ROC curves below, we explore the effect of changing the threshold values in Eq. 4.



(a) Percentile at line cut through the reconstructed microbubble pair shown in Fig. 5(b).

(b) Percentile at line cut through the false source shown and pointed out in Fig. 5(b).

(c) Percentile at line cut through the reconstructed microbubble pair shown in Fig. 5(e).

(d) Percentile at line cut through the false source shown and pointed out in Fig. 5(e).

Figure 6. Uncertainty (shown with percentiles) for real and false sources.

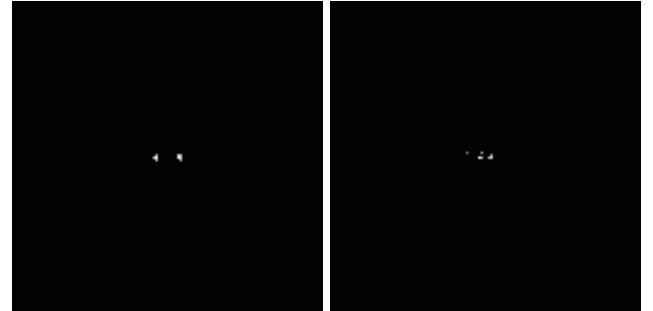


Figure 7. Filtering results by further filtering Fig. 5(b) and (e) (from left to right) with the simple percentile model $\mathcal{F}$.

Furthermore, it was shown in Fig. 6 that false sources have larger (normalized) variability than the real ones. To analyze this behavior statistically, multiple DAS images and their de-noised results were analyzed, as shown in Fig. 4, with real and false sources pointed out. In total, forty-four real sources and forty-five false sources are identified from randomly generated sample images. To modeling DAS pixel-wise magnitude distribution X , assuming the distributions are log-normal distribution, X can be written as

$$X = e^{\mu + \sigma Z}, \quad (5)$$

where Z denotes a standard normal distribution, μ and σ denote mean and standard deviation of the normal distribution. We solved for σ by fitting the pixel-level data to the log-normal model given by

$$\mathcal{N}(0, \sigma) = \sigma Z = \log X - \mu. \quad (6)$$

Fig. 8 shows the computed variances σ for all sources, along with their kernel density estimation for real and false sources, respectively. The standard deviation σ of the normal distribution $\mathcal{N}$ of the log-normal model is noticeably different for real and false sources. In most cases, false positive sources have larger σ in the log-normal model than real sources. This suggests that thresholding on the estimated σ provides another means of rejecting false positive sources. The performance is shown in Fig. 10.

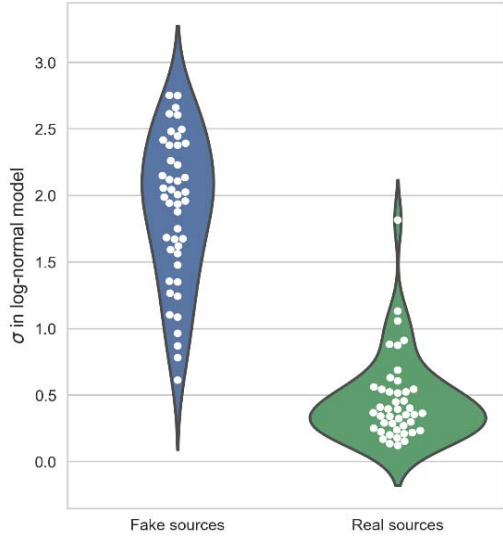


Figure 8. Estimated variabilities (σ) in log-normal model from Eq. 6, for the real and false sources.

Based on Eq. (6), we can also estimate both the log-scale averaged reconstructed magnitude as well as the variability σ . These two features can be used to establish a discriminant boundary for distinguishing fake sources. As an example, Fig. 9 shows a decision boundary generated using the support vector machine (SVM) algorithm using a linear kernel function.

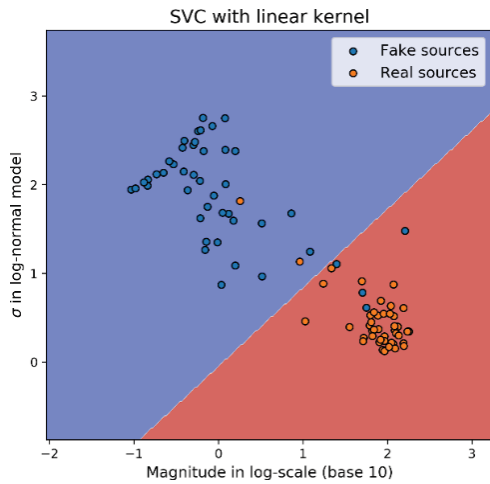


Figure 9. Distinguishing false positives with support vector machine.

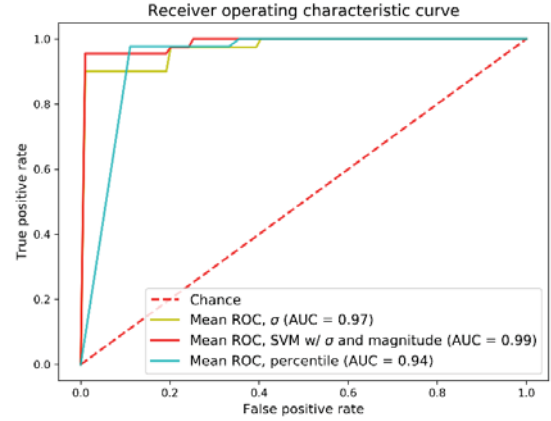


Figure 10. ROC curves of the real and false sources on σ model, on SVM model with σ and log-scale magnitudes, and on the percentile uncertainty analysis model, respectively. σ denotes the standard deviation of the normal distribution $\mathcal{Z}$ in the log-normal model.

ROC curves for the three proposed filtering steps are generated and shown in Fig. 10. The area under curve (AUC) is 0.99 found for applying SVM on σ in the log-normal model and log-scale magnitude, which is an improvement over the AUC of 0.97 found from the σ – only thresholding filtering model shown in Fig. 8, and the AUC of 0.94 found from the percentile uncertainty analysis model in Eq. 4.

V. DISCUSSION AND CONCLUSION

The U-Net-based method proposed here is generally successful in de-noising images from PCI and localizing microbubbles when micro-bubbles are not densely distributed. The noise reductions vs. DAS images range from 10-25dB. Although false sources are seen in the U-Net output, creating false positives as the micro-bubbles' distributions become denser, the information provided by MC dropout in the neural network provides additional cues for separating true and false sources. Thus, statistical tests are proposed which exploit the observation that true sources typically have higher amplitude and lower normalized variability than false sources. The larger normalized variability reflects higher uncertainty in the false sources. By further applying the log-normal filtering model on reconstructed sources, a high fraction of false sources can be distinguished and filtered out. Detailed examination of several of the remaining misclassified cases shows that they result from the complicated interference patterns that can occur when true sources are located very close to each other. We hypothesize that these errors can be further reduced by larger U-net training sets that include more examples of nearby sources. In future work, we plan to demonstrate the method on experimental data and to explore additional ways of exploiting uncertainty quantification information in deep learning outputs.

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